TAMID QUANT — STRATEGY ONE PAGER

Kalshi Longshot Tails Across BTC, S&P 500, Gold, WTI and Gas

Date: November 2026 | Presenter: Marcos Garay

The Idea

Kalshi's daily "above $K" contracts overprice longshots near expiry. On BTC alone the tail strategy earns many small wins and rare 3% losses, but trades only about half of days. We ran the same rule on the S&P 500, gold, WTI and natural gas ladders, whose big moves rarely coincide, and sized all markets together. On the Jul to Nov 2026 holdout, pooling across five markets made 4.55 winning trades a day against 1.82 for BTC alone, but no setting beat BTC alone on Sharpe by a significant margin.

Mathematical Framework

Core Model: *For market m with positions q_i (side s_i, fill p_i), the payoff if it settles in scenario j is*

$\Pi_{mj} = \Sigma_{i \text{ in } m}\, q_i \cdot [s_i \cdot (1[S_j > K_i] - p_i)]$ − fees, with one scenario between each pair of strikes. Market m's worst-case loss is $w_m = \max(0, -\min_j \Pi_{mj})$, exact rather than simulated.

Signal Construction: *Walk-forward YES rate per (horizon, price bin), prior expiry days only:*

S (separate): $f = (\Sigma y_m + N_0 \cdot a) / (n_m + N_0)$, $N_0 = 50$, where a is the bin's average price. **P (pooled):** $f_m = (\Sigma y_m + N_1 \cdot f_{pool}) / (n_m + N_1)$, $N_1 = 100$, where f_{pool} is the same estimate over all markets, so a new market starts from the cross-market bias.

Entry / Exit Logic: *6h and 3h before each market's own expiry, fill one hour later:* Mid < 10c or > 90c, spread ≤ 6c, edge ≥ 0.5c after 2c slippage and fee, 100 prior observations in the cell (pooled for P), 30-day warm-up per market. Hold to settlement.

Position Sizing: *One linear program per decision time, across markets:* maximize $\Sigma\, q_i \cdot edge_i$ subject to $w_m \le 3\%$ of bankroll for every m, and the two largest w_m summing to ≤ 4% (linear form: $2t + \Sigma u_m \le 4\%$, $u_m \ge w_m - t$, $u_m \ge 0$), $0 \le q_i \le 50$. Earlier fills enter as fixed positions. All five losing at once is capped near 10%.

Key Assumptions

- Each market has a longshot bias that past expiries reveal; nothing after the decision is used
- Big moves rarely hit more than two markets on the same day
- Hourly-candle quotes are fillable (no depth data); fee $0.07 \cdot q \cdot P \cdot (1 - P)$ per order, as Kalshi's API lists for all five series

Key Risks

- No proven edge: every holdout Sharpe gain over BTC alone has a 95% CI including zero; 14 settings compared, and the holdout was seen in earlier BTC work
- Gold, WTI and gas were listed in 2026 (116 to 128 expiry days), below the 150-day rule: Run B is exploratory
- WTI lost $19.73 in the holdout under pooling, on both sides of the book
- A shock moving three or more markets breaks the two-market cap (worst case about 10%)
- Fills come an hour after sizing; caps were exceeded by at most $0.76

At a Glance

Asset Class:	*Prediction Markets: Crypto, Equity Index, Commodities*
Strategy Type:	*Longshot Bias / Tail Risk Premium, Multi-Market*
Direction:	*Short Low-Tail YES, Long High-Tail YES*
Holding Period:	*3-6 Hours (Hold to Expiry)*
Universe:	*KXBTCD, KXINXU, KXGOLDD, KXWTI, KXNATGASD*

Backtest Results (holdout)

Holdout setting	Sharpe	Return	Wins / day
BTC only	1.07	+3.93%	1.82
A · S separate	0.86	+3.37%	1.96
A · P pooled	1.24	+5.98%	2.56
B · S separate	1.66	+5.47%	2.13
B · P pooled	0.67	+3.79%	**4.55**

Jul 1 to Nov 11 2026, $1,000 bankroll. A = BTC + S&P 500 (spec run); B = all five markets (exploratory). No Sharpe gain over BTC alone is significant.

Data Requirements

- Kalshi hourly bid, ask and settlement per contract - public API
- Expiry days: BTC 815, S&P 434 (from Feb 2025), gold 128, WTI 122, gas 116 (2026); about 72,000 quotes at 6h and 3h
- Order-book depth (not yet collected)

Key Parameters (frozen Nov 12 2026)

- Per-market cap 3%; two worst markets 4%; $1,000 bankroll
- Shrinkage $N_0 = 50$, pooling $N_1 = 100$; min 100 obs per bin; 30-day warm-up
- Tails: mid < 10c or > 90c; min edge 0.5c; slippage 2c; max 50 contracts

Open Questions

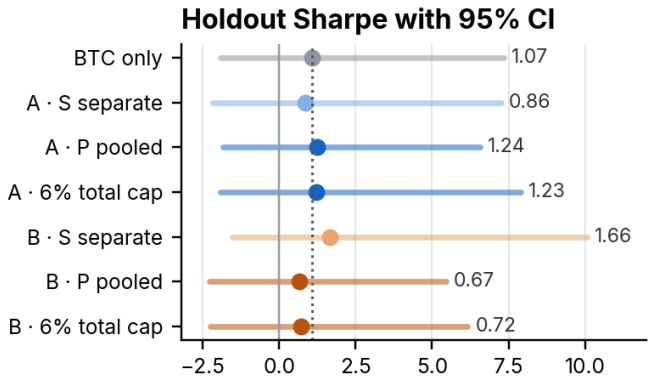
- Does the result hold on new data (forward test after Nov 11 2026)?
- Do implied volatility (VIX, GVZ, OVX) and event-day flags (FOMC, CPI, EIA) improve the estimates?
- Is there depth at 1c / 99c in the S&P and commodity books?
- Is WTI's loss real, or too little history?

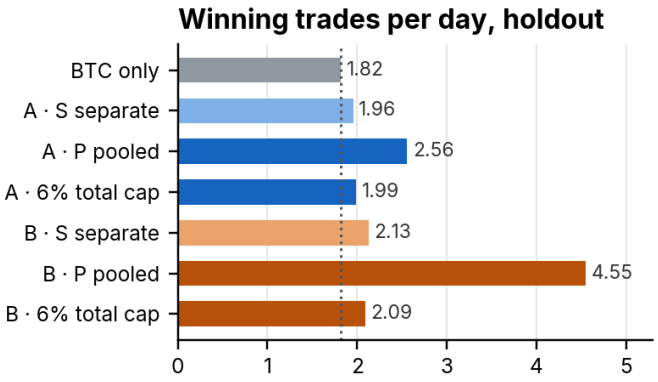
TAMID QUANT — STRATEGY RESULTS

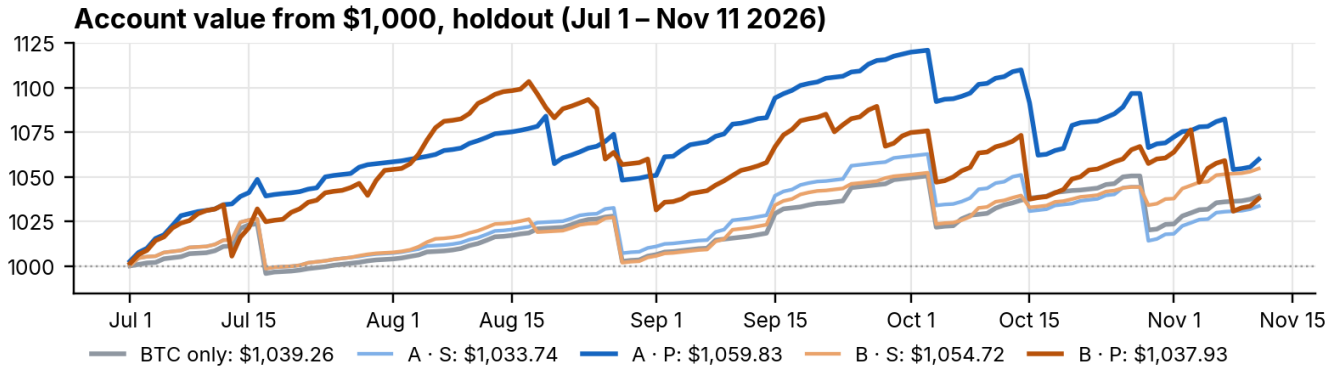
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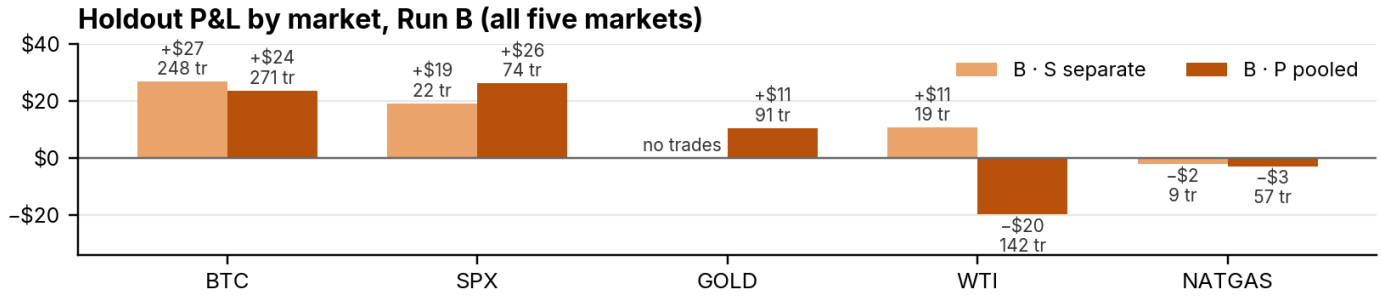
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Results on the holdout (Jul 1 to Nov 11 2026). Run A follows the spec (BTC + S&P 500); Run B adds gold, WTI and natural gas. S learns each market alone; P pools across markets. Each market's worst case is capped at 3% of a $1,000 bankroll and the two worst markets together at 4%.









Verdict

- **More wins: yes.** Pooling across five markets more than doubles winning trades per day (4.55 vs 1.82), and daily P&L between markets is close to uncorrelated (largest pair: S&P and WTI, 0.32). No day had two markets lose more than 1% of bankroll together
- **Better risk-adjusted return: not shown.** Every Sharpe interval overlaps BTC alone. More markets add more small wins but also more tail days
- **The S&P 500 is the addition to keep.** Under pooling it made $125.30 over its history (Run A), mostly from selling longshot YES. WTI is the weak market
- **Next:** settings are frozen for a forward test on data after Nov 11 2026, and a second-generation estimator (implied volatility, event days) will be judged on that same new data